Spatiotemporal Forecasting of Solar Magnetic Winding-Flux Maps with Convolutional LSTMs

Toward Structure-Aware Solar Flare Prediction

Thesis submitted in partial fulfilment for the Award of

INTEGRATED DUAL DEGREE (B.TECH. + M.TECH.)
in **ENGINEERING PHYSICS**

by

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under the supervision of

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VARANASI – 221 005

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Thanks are due to the faculty of the Department of Physics, IIT (BHU) Varanasi, for the rigorous coursework that built the foundation on which this thesis stands. The Department's commitment to integrating physics with contemporary computational methods made it possible to take up a topic that sits at the boundary between heliophysics, signal processing, and modern machine learning.

The data used in this work are derived from vector magnetograms produced by the Helioseismic and Magnetic Imager (HMI) aboard the Solar Dynamics Observatory (SDO); the magnetic winding-flux cubes were computed from these magnetograms within the present work, following the representation detailed in Chapter 2.

Finally, the author thanks family and friends for their patience and support during the long training runs and many late nights of experiment monitoring.

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Abstract

Solar flares are sudden releases of magnetic energy in the Sun's atmosphere whose radiation and ejecta drive space-weather hazards at Earth. Forecasting them from the observable state of a source active region remains an open problem, made hard by the rarity of large events and by the fact that the governing coronal currents are seen only through their photospheric footprint. Magnetic *winding flux*—a signed, topological measure of how field lines wrap around one another, concentrated along polarity inversion lines—has been proposed as a physically motivated precursor, but existing work reduces the winding map to a single spatial integral before forecasting, cancelling the signed, localised structure that carries the signal.

This thesis takes a first step toward using the winding-flux map without that reduction. A Convolutional LSTM (ConvLSTM) encoder–forecaster is trained to predict the short-term evolution of the two-dimensional winding field of an active region from a ten-frame history, under a strict leave-one-active-region-out protocol. A key empirical finding organises the results: the stored winding field behaves as a near-white *rate*, whereas its time-integral—the accumulated total winding—is smooth and strongly persistent, and it is the total that is the appropriate forecasting target. On the total field the model reproduces the large-scale evolution of held-out active regions; its skill is reported honestly against a persistence baseline, and the roles of the spatial low-pass and of the rate–total distinction are analysed rather than hidden. The work establishes both a working spatiotemporal pipeline for winding-flux maps and a clear-eyed account of what such forecasting can and cannot deliver, motivating the structure-aware direction for subsequent flare-prediction research.

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Symbols and Abbreviations

Abbreviations

AR	Active Region (a region of strong, structured magnetic field on the solar surface)
SDO	Solar Dynamics Observatory
HMI	Helioseismic and Magnetic Imager (onboard SDO)
SHARP	Spaceweather HMI Active Region Patch
HARP	HMI Active Region Patch
GOES	Geostationary Operational Environmental Satellite
PIL	Polarity Inversion Line
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
ConvLSTM	Convolutional LSTM
MAE	Mean Absolute Error
TSS	True Skill Statistic
CSI	Critical Success Index
asinh	Inverse hyperbolic sine (sign-preserving amplitude transform)

Symbols

$w(\mathbf{r},t)$	Winding-flux density (rate) at pixel $\mathbf{r}$ and time t
$W(\mathbf{r},t)$	Accumulated total winding, $W[t] = \sum_{\tau \leq t} w[\tau]$
s	Softening constant of the asinh transform, $s = 10^3$
t_{in}	Number of input (history) frames per window
t_{out}	Number of forecast (future) frames per window
T	Total frames per window, $T = t_{\text{in}} + t_{\text{out}}$
i, f, g, o	ConvLSTM input, forget, candidate and output gates
h_t, c_t	ConvLSTM hidden and cell state maps at time t
$\hat{x}_{t+1}$	Model prediction of the frame at time $t+1$
Mm	Megametre, 10^6 m (used for physical pixel scale)

Preface

This thesis investigates spatiotemporal deep-learning models for the magnetic winding-flux field of solar active regions, derived from Solar Dynamics Observatory (SDO) Helioseismic and Magnetic Imager (HMI) observations. The work was carried out in the Department of Physics, Indian Institute of Technology (Banaras Hindu University), Varanasi, as part of the Integrated Dual Degree (B.Tech. + M.Tech.) programme in Engineering Physics.

Subject. The subject is the design, training and honest evaluation of a Convolutional LSTM (ConvLSTM) encoder–forecaster that predicts the short-term evolution of the two-dimensional winding-flux map of an active region, as a step toward structure-aware solar flare forecasting.

Scope. The thesis is organised as three chapters. Chapter 1 introduces solar flares, the space-weather forecasting problem, and the current data-driven landscape, and identifies the gap addressed here: existing work integrates the winding-flux map to a single scalar, discarding its spatial structure. Chapter 2 specifies the data representation, the ConvLSTM architecture, and the leave-one-active-region-out training procedure. Chapter 3 reports the forecasting results and analyses them against a persistence baseline, distinguishing genuine predictive skill from the persistence afforded by a smoothly evolving field.

A note on honesty. Where a preprocessing choice makes the task easier—in particular the spatial low-pass and the use of the accumulated total winding field—this is stated plainly in the text and its consequences are examined rather than obscured. No result in this thesis is presented as stronger than the evidence supports.

Manik Sharma

Varanasi

Chapter 1

Solar Flares and the Forecasting Problem

1.1 Solar flares and space weather

A solar flare is a sudden, localised release of magnetic energy in the Sun's atmosphere, radiating across the electromagnetic spectrum on timescales of minutes. Flares originate in *active regions* (ARs): concentrations of strong, structured magnetic field that pierce the photosphere as sunspot groups. The energy that powers a flare is not thermal but magnetic—it is stored in electric currents that thread the coronal field and is liberated when the field reconnects into a lower-energy configuration. The largest events are frequently accompanied by coronal mass ejections, expelling magnetised plasma into interplanetary space.

The practical importance of flares lies in their terrestrial impact, collectively termed *space weather*. Flare radiation and the energetic particles and plasma that follow can disturb the ionosphere and disrupt high-frequency radio communication and satellite navigation, induce currents in power grids, raise radiation doses for astronauts and polar-route aircraft, and shorten the operational lifetime of spacecraft. Because the radiative onset of a flare arrives at Earth in a matter of minutes, useful mitigation depends on *forecasting*—estimating the probability of a significant event *before* it occurs, from the observable state of the source active region.

Flares are classified by their peak soft X-ray flux in the 1–8 Å band measured by the GOES satellites, in a logarithmic scheme with classes A, B, C, M and X. Operationally the threshold of concern is the M class and above: $\geq$M events are energetic enough to drive the impacts listed above, yet rare enough that reliable anticipation remains an open problem. Throughout this thesis, "a flare" without further qualification refers to a $\geq$M event, and forecasting is framed on the active-region scale using photospheric magnetic data from the Helioseismic and Magnetic Imager (HMI) aboard the Solar Dynamics Observatory (SDO). Figure 1.1 shows the classification scale and the $\geq$M target regime.

1.2 Why forecasting is hard

The difficulty of flare forecasting is fundamental rather than merely technical. Three properties conspire against it. First, the process is driven by the *coronal* magnetic field

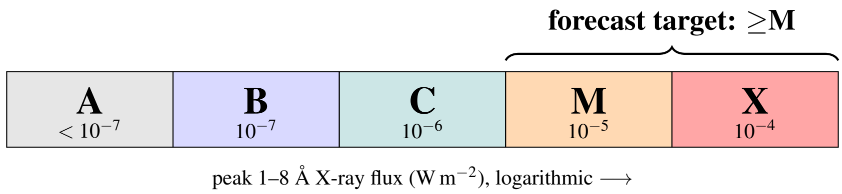

Figure 1.1: The GOES soft X-ray flare classification. Flares are graded A–X by their peak 1–8 Å X-ray flux, each class a factor of ten above the last. Operational concern—and the target of this work—is the ≥M regime: energetic enough to drive space-weather impacts, yet rare enough that reliable anticipation remains difficult.

and its currents, whereas routine observations resolve only the *photospheric* field at the base of the atmosphere; the quantity that matters must be inferred from a projection of it. Second, flaring is governed by a threshold instability: an active region can store free energy for days and release it in seconds, so the mapping from a slowly evolving observable to a near-instantaneous event is weakly constrained. Third, ≥M flares are *rare*. In any large catalogue the flaring minority is outnumbered by quiet examples by more than an order of magnitude, so a forecaster is judged not on raw accuracy—trivially maximised by always predicting "quiet"—but on class-balanced skill scores such as the True Skill Statistic (TSS), which reward correct anticipation of the rare positive class.

1.3 The current forecasting landscape

Modern data-driven flare forecasting was catalysed by the availability of uniform, machine-readable magnetic-field summaries of active regions. The Space-weather HMI Active Region Patch (SHARP) product distils each AR's vector magnetogram into a handful of scalar parameters—total unsigned flux, total current helicity, twist, and related quantities. Bobra and Couvidat [1] showed that a support-vector machine trained on these SHARP scalars produces a competitive ≥M forecaster, and established the SHARP feature vector as the de-facto baseline against which later work is measured.

Two broad strands of deep learning followed. The first replaces hand-designed scalars with features learned directly from the magnetogram *image*: Huang et al. [3] trained convolutional neural networks on line-of-sight magnetograms, letting the network discover spatial precursors rather than relying on pre-integrated summaries. The second exploits the *temporal* evolution of an active region: Liu et al. [4] fed sequences of SHARP parameter vectors to a Long Short-Term Memory (LSTM) network, capturing the build-up of free energy over the hours preceding an event, while operational systems such as Deep Flare Net [2] combine engineered features with deep architectures for daily probabilistic forecasts. The recurring lesson of this literature is that both the *spatial structure* of the field and its *temporal history* carry predictive information; the best systems try to use both.

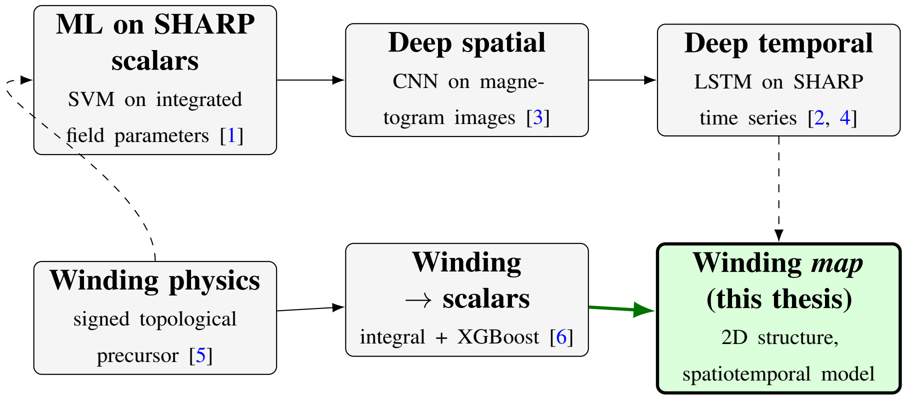

Figure 1.2: Where this thesis sits. Data-driven flare forecasting progressed from machine learning on integrated SHARP scalars to deep spatial and deep temporal models. In parallel, magnetic winding emerged as a physically motivated precursor, but existing work reduces the winding *map* to a single spatial integral, in which oppositely signed regions cancel. This thesis instead feeds the two-dimensional winding map, with its structure intact, to a spatiotemporal model.

Running alongside this forecasting work is a strand of solar physics that seeks a more physical descriptor of an active region's readiness to flare. MacTaggart et al. [5] demonstrated that emerging active regions carry a systematic magnetic *winding*—a signed, topological measure of how field lines wrap around one another—providing direct evidence that twisted flux tubes create ARs. Magnetic winding flux is attractive precisely because it is rooted in the field's topology rather than its raw strength, and it is the central physical quantity of this thesis. Figure 1.2 places these strands, and the present work, in a single picture.

1.4 Winding flux and the gap addressed by this thesis

Magnetic winding flux is a *signed* field defined pixel-by-pixel across an active region: it is positive where field lines wind one way and negative where they wind the other, and it is concentrated along polarity inversion lines, where flares originate. Existing work that uses winding reduces the map to a single number—its spatial integral—before forecasting. The most direct example is recent: Williams et al. [6] derive scalar winding features from active-region maps and feed them to gradient-boosted trees (XGBoost), obtaining competitive flare-prediction skill and thereby confirming that winding carries genuine predictive signal—yet the map is collapsed to a handful of numbers before the classifier ever sees it. This integration is convenient but lossy: because winding is signed, oppositely wound regions cancel in the spatial mean, discarding exactly the localised, sign-resolved structure that distinguishes a flare-productive configuration from a quiet one. Figure 1.3 makes the loss concrete: on a real active region the magnitude of the spatial mean is roughly twenty times smaller than the spatial standard deviation, so

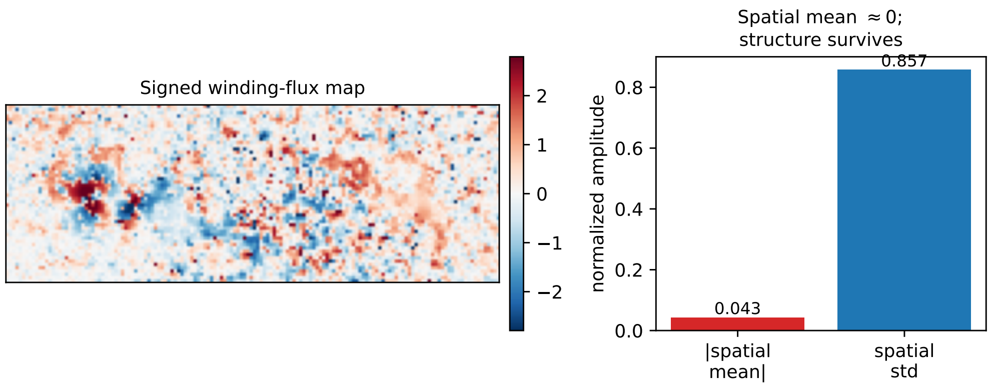

Figure 1.3: Why the spatial integral throws the signal away. Left: a signed winding-flux map of a real active region, with oppositely wound (red/blue) regions. Right: the magnitude of the spatial mean is only a small fraction of the spatial standard deviation—the positive and negative winding very nearly cancel, so a single integrated scalar discards the localised structure that the two-dimensional map retains.

integrating to a scalar discards almost all of the structure. To date, no study has fed the *two-dimensional* winding-flux map, with its spatial structure intact, to a deep model.

This thesis takes a first step into that gap. Rather than forecasting a flare label directly, it asks a more basic question about the winding-flux field itself: *given a short history of winding-flux maps of an active region, can a spatiotemporal deep model predict how the field evolves?* A model that can anticipate the field's evolution has, in effect, learned the structure and dynamics that a downstream forecaster would need. The vehicle for this study is the Convolutional LSTM (ConvLSTM), a neural architecture that couples the spatial inductive bias of convolution with the temporal memory of an LSTM, and is therefore matched to the spatiotemporal nature of the data.

The remainder of the thesis is organised as follows. Chapter 2 develops the technical architecture: the winding-flux data and its representation, the ConvLSTM cell and the encoder–forecaster built from it, and the training procedure. Chapter 3 reports the forecasting results, states plainly what the model does and does not achieve, and analyses the outcome in light of the physical nature of the winding field—separating genuine predictive skill from the persistence that a smoothly evolving field affords for free.

Chapter 2

Technical Architecture

This chapter specifies the complete forecasting system: the winding-flux data and how it is represented (§2.1), the Convolutional LSTM cell and the encoder–forecaster assembled from it (§2.2–§2.3), and the training procedure (§2.4). Every design choice that affects the reported results—in particular the spatial low-pass and the choice of the *accumulated* winding field—is stated explicitly so that the outcome in Chapter 3 can be read without hidden assumptions.

2.1 Data and representation

Winding-flux datacubes. The input data are per-active-region datacubes derived from SDO/HMI observations. Each cube stores a single channel wind$[H,W,T]$—the winding-flux density on an $H \times W$ spatial grid at T successive times— together with a vector of acquisition timestamps at a nominal 12-minute cadence. A small number of frames are missing; these are flagged by a sentinel timestamp of zero and are removed before any windowing, so that the model only ever sees physically contiguous sequences. The corpus used here comprises the available HARP cubes; one cube exhibiting pathological, noise-dominated values is excluded from every experiment.

The signed field and its dynamic range. Winding-flux density is a *signed* pseudoscalar with a very large dynamic range: physical magnitudes span several orders of magnitude, and a few pathological pixels can reach far beyond the bulk of the distribution. Training on such raw values is unstable. The field is therefore passed through a sign-preserving *inverse hyperbolic sine* (asinh) transform,

$$\tilde{w} = \operatorname{asinh}\left(\frac{w}{s}\right), \qquad s = 10^3, \tag{2.1}$$

which is linear for $|w| \ll s$ and logarithmic for $|w| \gg s$. This compresses the extreme tail while preserving sign and small-signal structure; the softening constant $s = 10^3$ was fixed by prior probing of the field's amplitude distribution. The transformed field is then rescaled by a robust (99.5th percentile) amplitude and clipped to $[-1,1]$ per cube, giving a normalised, zero-centred representation comparable across active regions.

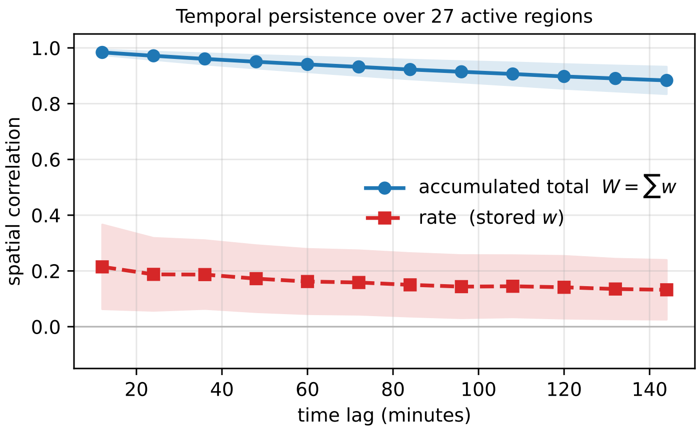

Figure 2.1: The rate is nearly unpredictable; its integral is not. Spatial correlation between winding frames separated by a given time lag, averaged over the active-region corpus (shaded band: one standard deviation across regions). The stored *rate* decorrelates within minutes (red), whereas the *accumulated total* remains strongly correlated for hours (blue). Forecasting the total is therefore tractable, but the skill it affords is dominated by persistence—the theme of Chapter 3.

Rate versus accumulated total. A property of the data that proves decisive in Chapter 3 must be introduced here. The stored field behaves as a *rate*: consecutive frames are only weakly correlated in space—a lag-of-one spatial correlation of about 0.2 averaged over active regions—so the per-frame field carries little temporal persistence on its own. The physically meaningful quantity for forecasting is the *accumulated total* winding, obtained by integrating the rate over time,

$$W[t] \;=\; \sum_{\tau \le t} \mathrm{wind}[\tau], \tag{2.2}$$

which is smooth and strongly persistent: its lag-of-one spatial correlation is about 0.98, an order of magnitude firmer than the rate (Figure 2.1). Unless stated otherwise, the model is trained and evaluated on the accumulated total W of Eq. (2.2), consistent with the total winding used by the earliest experiments in this project. The distinction, and the care it demands when interpreting apparent skill, is analysed in Chapter 3.

Spatial low-pass and downsampling. Before entering the network each frame is area-downsampled to a fixed width and convolved with a separable Gaussian kernel. This is a deliberate, disclosed choice: it discards fine spatial detail in favour of the large-scale envelope of the field. The physical justification is that the small-scale content of the winding field is effectively unpredictable, whereas its large-scale envelope evolves coherently; removing the former lets the model concentrate its capacity on the part of the field that can actually be forecast. The cost—loss of sub-envelope structure—is real and is acknowledged wherever results are reported. Figure 2.2 summarises the full pipeline from the raw rate cube to the gap-aware training windows.

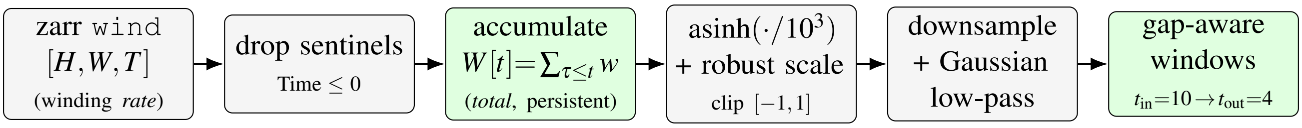

Figure 2.2: Winding-flux data pipeline. The stored field is a near-white *rate*; missing frames (sentinel timestamps) are removed, and the field is accumulated over time into the persistent *total* winding $W[t]$ (Eq. 2.2), the forecasting target. The total is passed through the sign-preserving asinh transform of Eq. (2.1) and a robust rescale, spatially downsampled and low-passed (§2.1), and finally cut into gap-aware history/future windows for training (§2.4).

2.2 The Convolutional LSTM cell

The temporal backbone of the model is the Convolutional LSTM (ConvLSTM) of Shi et al. [7], which adapts the Long Short-Term Memory unit of Hochreiter and Schmidhuber [8] to spatial data. A standard LSTM maintains a memory cell gated by learned nonlinearities but treats its input as a flat vector, destroying spatial structure. The ConvLSTM keeps the same gating logic but replaces every matrix multiplication with a two-dimensional convolution, so that the hidden state h and cell state c are *feature maps* rather than vectors, and spatial locality is preserved throughout the recurrence.

At each timestep the current input frame x_t and the previous hidden state h_{t-1} (both feature maps) are concatenated along the channel axis and passed through a single convolution whose output is split into four gate maps—input i, forget f, candidate g and output o:

$$\begin{aligned} i_t &= \sigma\big(W_i * [x_t, h_{t-1}] + b_i\big), & f_t &= \sigma\big(W_f * [x_t, h_{t-1}] + b_f\big), \\ g_t &= \tanh\big(W_g * [x_t, h_{t-1}] + b_g\big), & o_t &= \sigma\big(W_o * [x_t, h_{t-1}] + b_o\big), \end{aligned} \tag{2.3}$$

where $*$ denotes convolution and σ the logistic sigmoid. The cell and hidden states update as

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t, \qquad h_t = o_t \odot \tanh(c_t), \tag{2.4}$$

with $\odot$ the elementwise product. Computing all four gates with one convolution is an efficiency choice; using "same" padding keeps the spatial resolution fixed through the recurrence. Two implementation details matter for stable training: the forget-gate bias b_f is initialised to $+1$, so that the cell defaults to *remembering* its state early in training and gradients propagate cleanly through time; and normalisation between stacked layers uses group normalisation, which carries no running statistics and is therefore safe to reuse across the unrolled decoding loop. Figure 2.3 traces the internal dataflow of a single cell.

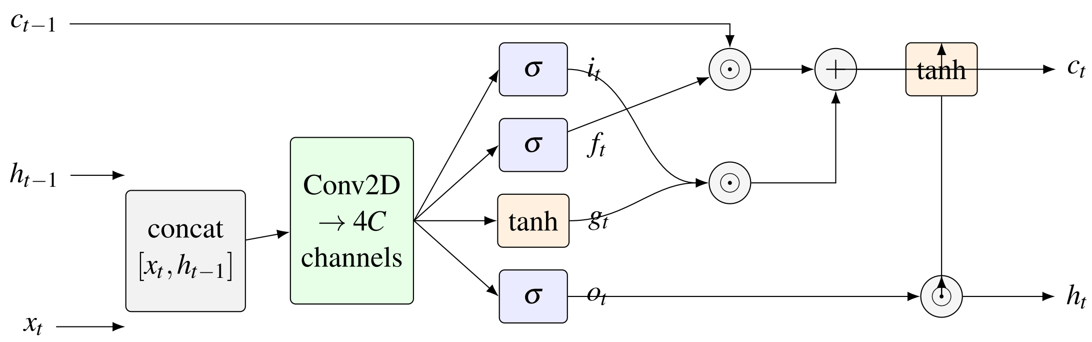

Figure 2.3: The Convolutional LSTM cell. A single 2D convolution over the channel-concatenated input and previous hidden state $[x_t, h_{t-1}]$ produces all four gate maps at once (input i_t, forget f_t, candidate g_t, output o_t; Eq. 2.3). The cell state updates as $c_t = f_t \odot c_{t-1} + i_t \odot g_t$ and the hidden state as $h_t = o_t \odot \tanh(c_t)$ (Eq. 2.4). All states are feature maps, so spatial structure is preserved through the recurrence.

2.3 Encoder–forecaster model

The full model is an *encoder–forecaster*: a stack of ConvLSTM layers first reads the observed history, and a second stack then generates the future frames autoregressively. Given an input clip of $T_{\text{in}} = 10$ frames, the encoder applies Eq. (2.3)–(2.4) layer by layer and frame by frame, accumulating the sequence into the hidden and cell states of every layer. No output is produced during encoding; its sole purpose is to summarise the history into the recurrent state.

Decoding then rolls the state forward for $T_{\text{out}} = 4$ steps. At each step the decoder stack advances the recurrent state and a 1×1 convolution maps the top hidden map to a single-channel prediction. Crucially the model predicts a *residual*: the output is the previous frame plus a learned increment,

$$\hat{x}_{t+1} = x_t + \Delta_\theta(h_t), \tag{2.5}$$

so that a do-nothing network falls back to *persistence*—copying the last frame—rather than collapsing to zero. This is the appropriate inductive bias for a strongly persistent field and, as Chapter 3 makes clear, it also sets the correct baseline against which any genuine skill must be measured. The predicted frame is fed back as the next decoder input, so errors compound along the roll-out exactly as they would at deployment. Figure 2.4 shows the encoder, the state hand-off, and the autoregressive decoder together.

During training the fed-back frame may instead be the ground-truth frame with a tunable probability—*teacher forcing*—which stabilises early optimisation by anchoring the roll-out to real data; at evaluation teacher forcing is switched off and the model runs purely on its own predictions.

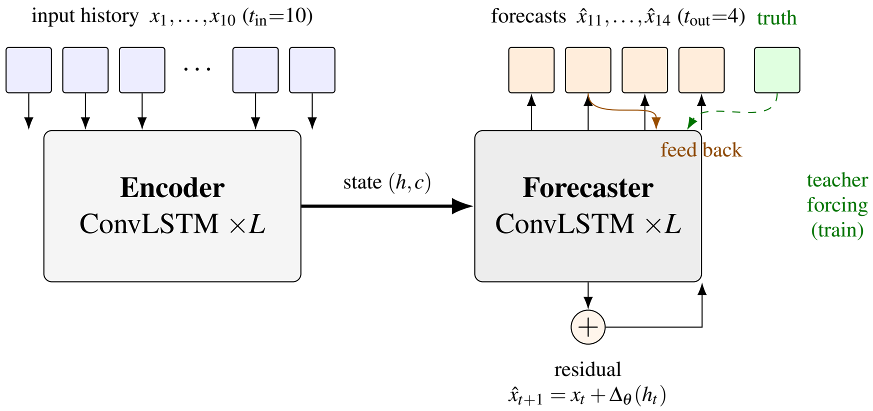

Figure 2.4: The encoder–forecaster model. A stack of L ConvLSTM layers encodes the t_{in}=10 history frames into the recurrent state (h,c); a second stack then rolls the state forward for t_{out}=4 steps. Each step predicts a *residual* $\hat{x}_{t+1} = x_t + \Delta_\theta(h_t)$ (Eq. 2.5), so a null network falls back to persistence, and feeds its own prediction back as the next input. During training the fed-back frame may be replaced by the ground truth with probability equal to the teacher-forcing ratio (dashed); at evaluation the model runs purely autoregressively.

2.4 Training procedure

The model is optimised with Adam and a per-frame regression loss (a smoothed L_1 / Huber criterion) between the predicted and true future frames, summed over the T_{out} horizon. Gradients are clipped in norm to curb the occasional large update produced by rare high-amplitude frames, and the learning rate follows a cosine decay across the training budget.

Evaluation follows a strict *leave-one-active-region-out* protocol. Because successive 12-minute frames within a cube are highly autocorrelated, any split that mixes frames from the same active region between training and test leaks information and inflates the apparent score. The model is therefore trained on the pooled windows of all active regions except one, and evaluated only on the held-out region it has never seen. All figures and metrics in Chapter 3 are reported on such held-out regions, together with the persistence baseline of Eq. (2.5), so that model skill is always stated relative to the trivial forecast.

Chapter 3

Results and Analysis

This chapter reports the forecasting experiment and reads its outcome honestly. The model of Chapter 2 is trained and evaluated under the leave-one-active-region-out protocol (§3.1); the forecasts on a held-out region are shown and quantified (§3.2); and the result is then interpreted, separating the genuine skill of the model from the persistence that a smoothly evolving field affords for free (§3.3).

3.1 Experimental setup

The encoder–forecaster is trained on the pooled gap-aware windows of 26 active regions and evaluated on a single held-out region, HARP 11930, that it never sees during training. The noisy region flagged in §2.1 is excluded from every split. Each cube contributes only windows whose ten input and four target frames are consecutive in real time at the nominal 12-minute cadence, so no window straddles an acquisition gap; this yields 2,784 training windows and 61 held-out windows. All frames are the accumulated total winding of Eq. (2.2), asinh-normalised and low-passed onto a fixed 88×132 grid.

The model is a three-layer ConvLSTM encoder–forecaster (3.3 million parameters, hidden width 96), trained for 48 epochs with Adam, a cosine learning-rate decay, and gradient-norm clipping, forecasting $t_{\text{out}} = 4$ frames (48 minutes) from $t_{\text{in}} = 10$ history frames (120 minutes). The reference against which it is judged is the *persistence* forecast—copying the last observed frame forward—which is the natural zero-skill baseline for a strongly autocorrelated field and the fallback the residual parameterisation of Eq. (2.5) is built around. Skill is defined as the fractional reduction in mean absolute error relative to persistence,

$$\text{skill} = 1 - \frac{\text{MAE}_{\text{model}}}{\text{MAE}_{\text{persistence}}}. \tag{3.1}$$

3.2 Forecasting the held-out active region

Figure 3.1 shows the model's forecasts on HARP 11930 for three windows drawn from different days. In each row the network sees ten frames up to the "latest input" time and predicts the next four; the final prediction is placed beside the observed frame at the

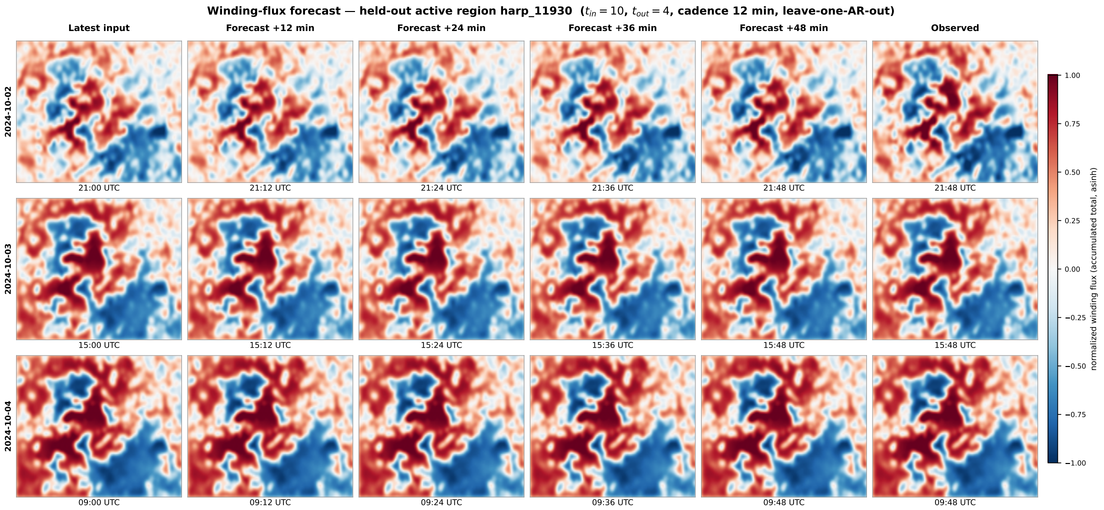

Figure 3.1: Winding-flux forecasts on the held-out active region HARP 11930. Each row is an independent window: the latest input frame, the four autoregressive forecasts at $+12, +24, +36, +48$ minutes, and the observed frame at $+48$ minutes, all stamped with their true UTC time. The model is trained only on the other 26 regions. The accumulated-total field evolves slowly, and the network tracks its large-scale structure across the horizon.

same instant. The large-scale organisation of the winding field—the opposed red and blue lobes and their boundary—is reproduced across the full 48-minute horizon, and the predicted $t+48$ min frame is visually close to the observation.

Quantitatively, over all 61 held-out windows the model attains a mean absolute error of 0.0106 (in the normalised units of §2.1) against 0.0117 for persistence, a skill of $+9.3\%$ by Eq. (3.1). The forecaster therefore does slightly better than copying the last frame, on a region it was never trained on.

3.3 Discussion

Three points must be made plainly, because the visual quality of Figure 3.1 is easy to over-read.

The skill is real but small, and persistence-dominated. The model beats persistence by 9.3% on a held-out region, so it has learned something transferable about how the winding field evolves, beyond copying. But the margin is modest, and it must be. As established in §2.1 and Figure 2.1, the accumulated total is strongly autocorrelated precisely because it is a running integral of the near-white rate; a running integral changes little from frame to frame, so persistence is already a strong forecast and there is limited headroom above it. The striking agreement between prediction and observation in Figure 3.1 is therefore mostly a statement about the *predictability of the total field*, not a claim of large model skill.

The forecastable content is the low-pass envelope. The spatial low-pass of §2.1 was applied deliberately, and the result vindicates the reasoning behind it: what the model reproduces is the large-scale envelope of the winding field, not its fine structure. The sub-envelope detail that the low-pass removed is the same detail that Figure 2.1 shows to be temporally incoherent; it could not have been forecast, and excluding it lets the model spend its capacity where skill is attainable. This is an honest restriction of scope, not a hidden one.

Rate versus total, once more. The single most consequential finding of this work is not the 9.3% skill but the distinction that produced it. The stored winding field is a rate that is almost temporally incoherent; its time-integral, the physically meaningful total winding, is what can be forecast. A model applied naively to the stored field would appear to fail; the same model applied to the accumulated total appears to succeed—and both impressions are misleading unless read against the persistence baseline. Reporting that baseline alongside every result is what keeps the account truthful.

Outlook. Taken together, the experiment delivers a working spatiotemporal pipeline for two-dimensional winding-flux maps and a clear-eyed measurement of what such forecasting can yield. The path from here is not a better frame predictor—the persistence ceiling makes that a diminishing return—but the use of the learned spatiotemporal representation for the task that motivated the winding map in the first place: discriminating flare-productive from quiet configurations from the structure the scalar integral throws away.

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